RECURRENT COMPACT QUANTUM GROUPS

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Abstract. We collect the important ideas of self-similar compact quantum groups in a non-technical report that aims to be accessible to those interested in studying them. Self-similar compact quantum groups are particular subsets of the quantum automorphisms of a infinite homogeneous rooted tree. We propose that recurrent compact quantum groups should be self-similar quantum group with restriction being, in some sense, injective.

1. Introduction

Self similar groups can be thought of as groups generated by automata, that is a system that starts in some state q_0 and letter-by-letter reads some string $a_1a_2...$ which updates the state q, and outputs some letter [3]. More formally, the action of a group G on the set of all $n-$ary strings X^* is said to be self similar if for each $g \in G$, all words $w \in X^*$, and all letters $x \in T$

$$g(xw) = g(x)g|_x(w)$$

for some $g|_x \in G$ known as the restriction of g by x [3]. Very simple automata were found to generate very interesting groups, one famous example is Grigorchuk's group, which was the first example of a group with intermediate growth and the first known amenable group to not be elementary amenable [2]. Further, self-similar groups have been shown to be connected to dynamics and fractal geometry [3]. A particular subclass of self-similar groups are *recurrent groups*, which require certain surjective conditions on the restriction map.

The study of symmetries has been a long standing part of mathematics. However, modern mathematics and particle physics requires the extension of symmetries to non-commutative spaces. This lead to the development of quantum groups, which have now become a significant part of modern mathematics. In some sense, quantum groups are the generalisation of the space of continuous functions $C(G)$ of a group G to an extent that allows for non-commutativity [8]. A substantial part of the study of quantum groups involves the study of Woronowicz's *compact quantum groups* [9].

Recent work by Brownlowe has answered the question of *"is there a suitable notion of self-similar groups that applies to compact quantum groups?"* [1]. In doing so, they established the importance of the quantum automorphisms of a rooted homogeneous tree $\mathbb{A}_X$, and developed a suitable definition for self-similar quantum subgroups. This work aims to clearly define the notions of self-similarity, compact quantum groups, and to demonstrate how one can naturally develop the definition established by Brownlowe. As this report is designed to be somewhat non-technical, a number of remarks are scattered through, explaining how one can intuitively grasp the ideas being discussed. Additionally, we aim to establish a definition of recurrent quantum groups.

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Section 2 introduces some preliminary definitions for C^*-algebras and tensor products. Section 3 establishes the idea of a self-similar and recurrent group in terms of the automorphisms of a rooted tree. Section 4 defines a compact quantum group, and the generalisation of some important ideas such as quantum subgroups. Section 5 defines $\mathbb{A}_X$ the quantum version of the infinite rooted homogeneous tree, in section 6 we use this to to define self-similarity and recurrence for compact quantum groups. We finish with a brief conclusion in section 7.

2. Preliminaries

Here we introduce the definition of a $C*-$algebra and the tensor product, we will not be making heavy use of these definitions, but they are here for completeness.

Definition 1. For R a ring, a *left $R-$module* is a pair $(M, \cdot)$ where M is an abelian group, and $\cdot : R \times M \to M$ such that for all $r, s \in R$ and $x, y \in M$ we have

- $r \cdot (x + y) = r \cdot x + r \cdot y$
- $(r + s) \cdot x = r \cdot x + s \cdot x$
- $(rs) \cdot x = r \cdot (s \cdot x)$
- $1 \cdot x = x$ where 1 is the identity of the ring R

The definition of a *right $R-$module* is obvious. If we let R be commutative, the left and right $R-$modules agree, and we simply call this an *$R-$module*.

Remark. An $R-$module can be viewed as the extension of a vector spaces that allows the scalars to form a ring instead of a field.

Definition 2. An *associative algebra* is an $R-$module for some commutative ring R, extended to be a ring in such a way that the addition of the module and the associated ring agree, and scalar multiplication satisfies

$$r \cdot (xy) = (r \cdot x)y = x(r \cdot y)$$

Remark. Associative algebras are like vector spaces where vectors can be multiplied.

Definition 3. A *normed vectors space* is a vector space V together with a norm $\| \cdot \| : V \to \mathbb{R}$ such that

- $\|x + y\| \leq \|x\| + \|y\|$
- $\|sx\| = |s|\|x\|$
- if $\|x\| = 0$ then $x = 0$

This norm induces a metric $d(x, y) = \|x - y\|$. So, we have a metric space.

Remark. Normed vector spaces are vector spaces where we can measure the length of vectors.

Definition 4. In a metric space, a sequence $x_1, x_2, ...$ is *Cauchy* if for all real $r > 0$, there exists some N such that $d(x_n, x_m) < r$ for all $m, n > N$. A normed vector space X is called *Banach* if for every Cauchy sequence, there is some $x \in X$ such that

$$\lim_{n\to\infty} x_n = x$$

Remark. In some sens, Banach spaces have no holes.

Definition 5. A *Banach algebra* is an associative algebra A that is also a Banach space.

Definition 6. A C^*-algebra A is a Banach algebra over $\mathbb{C}$ with an involution $* : x \mapsto x^*$ such that for all $x, y \in A$ and $\lambda \in \mathbb{C}$

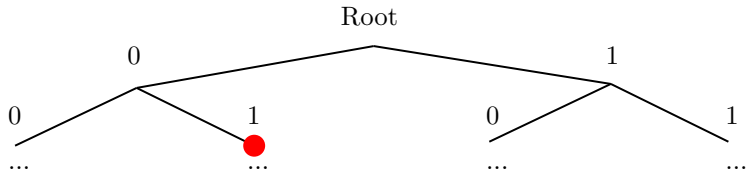


FIGURE 1. Infinite binary rooted homogeneous tree, highlighted node $w = 01$.

- $(x + y)^* = x^* + y^*$
- $(xy)^* = y^* x^*$
- $\|xx^*\| = \|x\|^2$

Remark. One can think of a $C*-$algebra as the generalisation of the space of complex matrices where we measure a matrix by looking at the maximum amount that it scales a vector ($\|M\| = \max |Mu| : |u| = 1$). It is a space with no holes in it, where we can multiply elements together, measure their length, multiply them by complex numbers, and take their adjoint.

The following is the universal definition of the tensor product, there are a number of other more constructive definitions, see for example Ryan [6].

Definition 7. For any two vector spaces V, W, their tensor product is another vector space denoted $V \otimes W$ and an associated map $\otimes : (v, w) \mapsto v \otimes w$ which is bilinear, such that for any other bilinear map $f : V \times W \to X$ there is a unique linear map $\tilde{f} : V \otimes W \to X$ such that $\tilde{f}(v \otimes W) = f(v, w)$.

Remark. For those unfamiliar with the tensor product, a suitable overview of the topic can be developed by substituting it with the Cartesian product.

Remark. This tensor product can be extended to the *spatial tensor product* of C^*-algebras by considering the embedding of the algebraic tensor product $V \otimes W$ in $B(\mathcal{H})$, the bounded functions of some Hilbert space. Appendix B of Raeburn and Williams provides details [5] . When referring to the tensor product of a C^*-algebra, we are referring to this spatial tensor product.

3. Self-Similarity

We denote by X^* the binary infinite rooted tree, as in figure 1. We can label the nodes of X^* by binary "words" $w = w_1 w_2 w_3 ....$ where each letter w_i represents which branch we must follow to get to the node. For example, the highlighted node in figure 1 is denoted by $w = 01$. Denote by $|w|$ the number of letters in a word, known as the length of the word.

Any bijective mapping from nodes to nodes of X^* that preserves adjacency's (that is words that are connected by a edge stay connected) we call an automorphism. We denote by $\mathrm{Aut}(X^*)$ the set of all automorphisms of X^*. If we have two words u, v we denote their concatenation (that is the word constructed by placing u next to v) by $w = u.v$, unless it is obvious what we are talking about, in which case we simply write $w = uv$. We have the following important property of automorphisms

Proposition 8. *For any automorphism $g \in \mathrm{Aut}(X^*)$, and words u, v there is a word w such that*

$$g(uv) = g(u).w \tag{1}$$

We can easily extend these idea to infinite rooted $n-$ary trees. We call the set of all letters of X^* the "alphabet" and denote it by X.

We have the following definitions

Definition 9. Let G be a group of automorphisms of X^*, we call G *self-similar* if for every letter $x \in X$, each word w, and each $g \in G$, there is a $y \in X$ and $h \in G$ such that

$$g(xw) = y.h(w) \tag{2}$$

Definition 10. The restriction of g by u is the unique mapping $g|_u$ such that for any word v

$$g(uv) = g(u).g|_u(v) \tag{3}$$

Proposition 11. *An important property of restriction is that*

$$g|u_1|u_2 = g|u_1u_2 \tag{4}$$

Remark. the proposition is very useful as it allows us to only study restrictions by words of length 1.

Proposition 12. *A group G is self similar if and only if it is closed under restrictions.*

Proof. ($\Longrightarrow$)

$$\begin{aligned} g(u_1u_2u_3...u_nv) &= y_1h_1(u_2u_3...u_nv) \\ &= y_0y_1h_2(u_2...u_nv) \\ &... \\ &= y_1y_2y_3...y_nh_n(v) \\ &= g(u_1u_2u_3...u_n)h_n(v) \end{aligned}$$

Where each $h_i \in G$ by definition 9. The last line comes from proposition 8. By comparison $h_n(v) = g|_u(v)$.

($\Longleftarrow$) By definition, $g(xw) = g(x)g|_x(w)$ with $g|_x \in G$, simply relabeling $g|_x$ by h and $g(x)$ by y gives us the definition of self similarity. □

Moving forwards, we will use closure under restrictions to define self similarity.

Definition 13. The *restriction map* is the map $\phi_x : G \to G$ defined by $\phi_x(g) = g|_x$.

Theorem 14. *The restriction map is uniquely characterised by the following properties.*

- $\phi_x(e) = e$
- $\phi_x(gh) = \phi_{h(x)}(g)\phi_x(h)$

All the above definitions extend to actions of groups on X^*. To construct a self-similar action of G on X^* it suffices to define how the elements of G act on X^*, and to define a restriction map with the properties of theorem 14. Hence we have the following

Corollary 15. *To have a self-similar action of G on X^*, it is sufficient to have a function $f : G \times X \to X \times G$ such that $f(e, x) = f(x, e)$ for all $x \in X$ and the following diagram commutes*

$$\begin{array}{ccc} G\times G\times X & \xrightarrow{m_G\times \mathrm{id}_X} & G\times X \\ \downarrow{\scriptstyle \mathrm{id}_G\times f} & & \\ G\times X\times G & & \downarrow{\scriptstyle f} \\ \downarrow{\scriptstyle f\times \mathrm{id}_G} & & \\ X\times G\times G & \xrightarrow{\mathrm{id}_X\times m_G} & X\times G \end{array}$$

Where $(g\cdot x, g|_x) := f(g,x)$.

Definition 16. A self-similar group G is *recurrent or fractal or self-replicating* if for every $x, y \in X$ and $h \in G$, there exists a $g \in G$ such that

(1) $g(x) = y$
(2) $\phi_x(g) = h$

The first condition is called transitivity on the first level, and the second condition encapsulates subjectivity of ϕ_x [4].

Remark. This definition essentially states that the restriction when applied to the subset of functions with $g(x) = y$ is surjective.

Definition 17. $G_{y,x} = \{g \in G : g(x) = y\}$

Recurrence is clearly equivalent to the statement that for all $x, y \in X$ and $h \in G$ there is some $g \in G_{y,x}$ (by condition 1) such that $\phi|_x(g) = h$ (by condition 2). This gives the following theorem.

Theorem 18. *A self-similar group G is recurrent if and only if for all $x, y \in X$, $\phi_x : G_{y,x} \to G$ is surjective.*

Remark. Here the map $\phi|_x : G_{y,x} \to G$ is simply $\phi|_x \circ i$ where $i : G_{y,x} \to G$ is the inclusion map and $\phi|_x : G \to G$ is the usual restriction map.

4. Compact Quantum Groups

Woronowicz gave use the following definition of a compact quantum group [9].

Definition 19. Let $G = (A, \Phi)$ where A is a C^* algebra, and $\Phi : A \to A \otimes A$ is a $*$-homomorphism called comultiplication. We say that G is a compact quantum group if the sets

$$\{(b \otimes I)\Phi(c) \ : \ b, c \in A\}$$

and

$$\{(I \otimes b)\Phi(c) \ : \ b, c \in A\}$$

are linearly dense subsets of $A \otimes A$, and the following diagram commutes

$$\begin{array}{ccc} A & \xrightarrow{\Phi} & A\otimes A \\ {\scriptstyle \Phi}\downarrow & & \downarrow{\scriptstyle \Phi\otimes \mathrm{id}} \\ A\otimes A & \xrightarrow[\mathrm{id}\otimes\Phi]{} & A\otimes A\otimes A \end{array}$$

We call this property coassociativity.

Remark. In this sense, comultiplication is a generalisation of the comultiplication of the continues functions of an algebra, $C(G)$ defined as $(\Phi f)(g,h) = f(gh)$, which in itself is the dual of multiplication.

Woronowicz also proved the following [9]

Theorem 20. *(A,Φ) is a compact quantum group if and only if there is a family of matrices $\left\{a^\lambda = (a^\lambda_{ij}) \in M_{d_\lambda}(A) : \lambda \in \Lambda)\right\}$ dense in A such that*

(1) $\Phi(a^\lambda_{ij}) = \sum_{k=1}^{d_\lambda} a^\lambda_{ik} \otimes a^\lambda_{kj}$ for all $\lambda \in \Lambda$ and $1 \leq i,j \leq d_\lambda$
(2) a^λ and $(a^\lambda)^T$ are invertible

Example 21. Take S_n to be the permutation group with elements σ. These elements σ have representations in the $n \times n$ matrices. For each i,j, let $f_{i,j} \in C(S_n)$ be defined as $f_{i,j}(\sigma) = \sigma_{i,j}$. These functions $f_{i,j}$ generate $C(S_n)$ and have defining properties that

(1) $f_{i,j} = f^*_{i,j} = f^2_{i,j}$
(2) $\sum_i f_{i,j} = 1 = \sum_j f_{i,j}$

We have comultiplication defined as

$$\Phi f_{i,j}(\sigma,\tau) = f_{i,j}(\sigma\tau) = \sum_k \sigma_{i,k}\tau_{k,j} = \left(\sum_k f_{i,k} \otimes f_{k,k}\right)(\sigma,\tau)$$

.

We hence define the quantum permutation group to be the universal C^* algebra generated by elements $a_{i,j}$ satisfying the same properties as above. A proof that this is indeed a quantum group can be found using theorem 20 [7].

Definition 22. An ideal I is a *Woronowicz ideal* if $\Phi(I) \subseteq \ker(q \otimes q)$ where q is the quotient map. A *quantum subgroup* of (A,Φ) is a pair $(A/I,\Phi')$, I a Woronowicz ideal, where Φ' satisfies $\Phi' \circ q = (q \otimes q) \circ \Phi$. This pair is indeed a quantum group.

Remark. The use of a Woronowicz ideal ensure that $\Phi'(0) = 0$.

Definition 23. A (left) *coaction* of a quantum group (A,Φ) on a C^* algebra B, is a $*-$homomorphism $\alpha : B \to A \otimes B$ where $\alpha(B)(1 \otimes A)$ is dense in $A \otimes B$ and the following diagram commutes

$$\begin{array}{ccc} B & \xrightarrow{\alpha} & A \otimes B \\ {\scriptstyle\alpha}\downarrow & & \downarrow{\scriptstyle\Phi\otimes\mathrm{id}} \\ A \otimes B & \xrightarrow[\mathrm{id}\otimes\alpha]{} & A \otimes A \otimes B \end{array}$$

Remark. The density condition is known as *Podle´s condition*. The diagram encapsulates the dual of the property that $g \cdot (h \cdot x) = gh \cdot x$ in

5. Quantum Automorphisms of an Infinite Rooted Tree

We begin by defining the action of a quantum group on X^*

Definition 24. The action of a quantum group (A,Φ) on X^* is a system of left coactions $\alpha = (\alpha_n : C(x^n) \to A \otimes C(X^n))$ such that the following diagram comutes

$$\begin{array}{ccc} C(X^n) & \xrightarrow{i_{m,n}} & X^n \\ \downarrow{\scriptstyle\alpha_m} & & \downarrow{\scriptstyle\alpha_n} \\ A \otimes C(X^m) & \xrightarrow{\mathrm{id}\otimes i_{m,n}} & A \otimes C(X^n) \end{array}$$

where $i_{m,n}$ is an injective homomorphism satisfying $i_{m,n}(p_w) = \sum_{w' \in X^{n-m}} P_{ww'}$

Remark. The above definition is the dual of proposition 8, with $i_{m,n}$ the dual of "chopping off" the end of a word.

As was done in 21, we examine the functions that generate $C(\mathrm{Aut}(X^*))$. These are the functions $f_{u,v} := 1_{g \in \mathrm{Aut}(X^*) : g(v) = u}$ for $u, v \in X^n$, which satisfy

$$f_{u,v} = \sum_{y \in X} f_{ux,vy} = \sum_{z \in X} f_{uz,vx} \text{ for any } x \in X$$

Comultiplication is defined as $\Phi(f_{u,v})(g,h) = f_{u,v}(gh)$ which is 0 unless $(gh)(v) = u$, in which case it is 1. This is equivalent to there existing some word w such that $h(v) = w$ and $g(w) = u$. So,

$$\Phi(f_{u,v}) = \sum_{|w| = |u| |v|} f_{u,w} \otimes f_{w,v} \tag{5}$$

Definition 25. $\mathbb{A}_X$ is the universal C^* algebra generated by elements $a_{i,j}$ such that

- $a_{\emptyset,\emptyset} = 1$
- $a_{u,v} = a_{u,v}^* = s_{u,v}^2$
- $a_{u,v} = \sum_{y \in X} a_{ux,vy} = \sum_{z \in X} a_{uz,vx}$ for any $x \in X$

And comultiplication is as in equation 5.

A proof that this is indeed a quantum group equivalent of $\mathrm{Aut}(X^*)$ is provided by Brownlowe [1].

6. Self-Similar Compact Quantum Groups

We start by examining the dual of the restriction map ϕ_w. This is defined as $\rho_w(f)(g) = f(\phi_w(g)) = f(g|_w)$. It is defined by its action of the elements $f_{u,v} = \mathbf{1}_{G_{u,v}}$ (as defined in definition 17) that generate $C(\mathrm{Aut}(X^*))$

$$\rho_w(f_{u,v})(g) = f_{u,v}(g|_w) \tag{6}$$

$$= \begin{cases} 1, & \text{if } g|_w(v) = u \\ 0, & \text{otherwise} \end{cases} \tag{7}$$

$$= \begin{cases} 1, & \text{if } g(xv) = yu \text{ for some } |y| = |w| \\ 0, & \text{otherwise} \end{cases} \tag{8}$$

$$= \sum_{|y| = |w|} f_{yu,wv}(g) \tag{9}$$

This extends obviously to compact quantum groups.

Proposition 26. *For each $w \in X^*$, there is a homomorphism $\rho_w : \mathbb{A}_X \to \mathbb{A}_X$ satisfying*

$$\rho_w(a_{u,v}) = \sum_{|y| = |w|} a_{yu,wv} \tag{10}$$

A proof of this is provided by Brownlowe [1].

Definition 27. We call ρ_w as in proposition 26 the *quantum restriction map by* w. A quantum subgroup A of $\mathbb{A}_X$ is *self-similar* if for each $x \in X$ there exists some $\tilde{\rho}_x : A \to A$ such that the following diagram commutes

$$\begin{array}{ccc} \mathbb{A}_X & \xrightarrow{\rho_x} & \mathbb{A}_X \\ \downarrow{\scriptstyle q} & & \downarrow{\scriptstyle q} \\ A & \xrightarrow{\tilde{\rho}_x} & A \end{array}$$

Remark. This definition is a consequence of the generalisation of corollary 15 [1]. It is essentially saying that ρ_x exists in A, except as A is a quotient of $\mathbb{A}_X$ we need to be careful and use the quotient map to establish ρ_x in A

To develop the quantum notion of a recurrent group (definition 16) we must examine a functional definition.

As outlined in theorem 18, a self-similar group G is recurrent if for all $x, y \in X$, the following composition of maps is surjective

$$G_{y,x} \xrightarrow{i} G \xrightarrow{\phi_x} G$$

We extend this definition to quantum groups by taking its dual.

Proposition 28. *A quantum subgroup A of $\mathbb{A}_X$ is recurrent if for all $x, y \in X$, the following composition of maps is injective*

$$A/I \xleftarrow[q]{} A \xleftarrow[\tilde{\rho}_x]{} A$$

where $I = \langle a_{z,x} \rangle$ such that $z \neq y$.

7. Conclusion

This work has collected many of the important results required to establish the definition of self-similar quantum groups. It has also extended this work to include a definition of recurrent quantum groups. Further work needs to be done in refining the definition of a recurrent quantum group, developing quantum definition of other important ideas from the study of self-similar groups, and finding more examples of self-similar and recurrent quantum groups.

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